

Predicting Online Shopper Purchase Intent

Using Machine Learning to Identify High-Intent E-Commerce Sessions



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Capstone Three

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Executive Summary

This project examined whether information about a visitor's browsing behavior and online shopping session could be used to predict whether the session would result in a purchase. Using the Online Shoppers Purchasing Intention Dataset containing 12,330 sessions, three machine learning approaches, namely Logistic Regression, Random Forest, and Deep Neural Networks were developed and compared. The Tuned Random Forest was selected as the final model because it provided the strongest overall balance between identifying purchasing sessions and maintaining predictive performance.

Key Findings

- 15.5% of online shopping sessions resulted in a purchase, creating a moderately imbalanced outcome.
- The Tuned Random Forest achieved the highest recall (68.3%), F1-score (0.663), and ROC-AUC (0.923) among all models.
- The Baseline Random Forest achieved the highest accuracy (90.0%) and precision (76.6%) but identified fewer purchasing sessions.
- PageValues was by far the most important feature, followed by ExitRates, product-related browsing activity, BounceRates, and administrative page activity.
- Overall, visitor engagement and browsing behavior were more useful for predicting purchase intent than broader visitor or temporal characteristics.
- The results suggest potential applications for identifying high-intent sessions and supporting targeted interventions, although further validation would be needed before real-world implementation.

1.0 Introduction

1.1 Why Does This Problem Matter?

Online shopping has become an important part of how consumers purchase goods and services, but not every visit to an e-commerce website results in a purchase. Customers may browse products, compare options, or leave without completing a transaction. For businesses, understanding which sessions are more likely to result in a purchase can provide useful information for improving customer experience and targeting marketing and engagement efforts.

Information about the pages a visitor views, the time spent browsing, and other characteristics of a shopping session may provide useful signals about purchase intent. Machine learning offers an opportunity to identify these patterns and develop models that distinguish between sessions that are more or less likely to result in a purchase.

This project examines whether online shopping behavior and session characteristics can be used to predict purchase outcomes. Multiple classification models are developed and compared with the final model used to identify the features that provide the most useful predictive information.

1.2 Project Context and Audience

This analysis demonstrates how machine learning can support decision-making in an e-commerce setting. The findings may be relevant to e-commerce business leaders, marketing and analytics teams, and product or user experience teams interested in understanding customer behavior and improving conversion.

1.3 Project Objectives

The objectives of this project were to:

1. Develop machine learning models to predict whether an online shopping session will result in a purchase.
2. Compare the performance of multiple machine learning algorithms using accuracy, precision, recall, F1-score, and ROC-AUC.
3. Identify the features that are most useful for predicting whether a browsing session results in a purchase.
4. Examine the potential business applications of the final model and discuss how purchase intent predictions could support decision-making in an e-commerce setting.

2.0 Data

2.1 Data Source

This project uses the Online Shoppers Purchasing Intention Dataset, which contains information on individual online shopping sessions. The dataset was created to examine whether characteristics of a user's browsing behavior during an online shopping session could be used to predict whether the session resulted in a purchase.

Each observation represents a single online shopping session. The available variables capture different aspects of the session, including the number of pages visited, time spent on different types of pages, bounce and exit rates, the month of the session, traffic source, visitor type, and whether the session occurred on a weekend.

The dataset contains 12,330 online shopping sessions and 17 predictor variables. The outcome variable, Revenue, indicates whether the session resulted in a purchase.

Table 1. Dataset Summary¹

Characteristic	Description
Dataset Name	Online Shoppers Purchasing Intention Dataset
Data Source	UCI Machine Learning Repository
Original Dataset/Source	Data collected from an online shopping website
Dataset Period	One year of online shopping sessions
Sample Size	12,330 individual online shopping sessions
Target Variable	Revenue
Outcome Definition	Whether the online shopping session resulted in a purchase

2.2 Outcome Variable

The outcome variable for this project was Revenue, which indicates whether an online shopping session resulted in a purchase.

The variable is boolean:

- False = No Purchase — the session did not result in a completed transaction.
- True = Purchase — the session resulted in a completed transaction.

¹ Sakar, C. and Yomi Kastro. "Online Shoppers Purchasing Intention Dataset." UCI Machine Learning Repository, 2018, <https://doi.org/10.24432/C5F88Q>.

Approximately 15.5% of sessions resulted in a purchase, creating a moderately imbalanced outcome. Because of this imbalance, model performance was evaluated using precision, recall, F1-score, and ROC-AUC in addition to accuracy.

2.3 Predictor Variables

The Online Shoppers Purchasing Intention dataset contains 17 predictor variables that describe visitor browsing behavior, website engagement, session characteristics, and visitor attributes. The variables are grouped into four categories: browsing behavior, website engagement and page-level metrics, timing and seasonal characteristics, and visitor and technical characteristics.

2.3.1 Browsing Behavior

These variables capture how visitors interact with different types of pages during their shopping session. The values are derived from the URLs visited during the session and are updated as the visitor takes actions, such as moving from one page to another.

Table 2. Browsing Behavior Variables

Variable	Description
Administrative	Number of administrative pages visited during the session.
Administrative_Duration	Total time spent on administrative pages during the session.
Informational	Number of informational pages visited during the session.
Informational_Duration	Total time spent on informational pages during the session.
ProductRelated	Number of product-related pages visited during the session.
ProductRelated_Duration	Total time spent on product-related pages during the session.

2.3.2 Website Engagement and Page-Level Metrics

These variables represent metrics measured by Google Analytics for pages visited during the shopping session. They provide additional information about visitor engagement and the relationship between pages viewed and eventual transaction activity.

Table 3. Website Engagement and Page-Level Variables

Variable	Description
BounceRates	The percentage of visitors who enter the site through a page and leave without triggering another request to the analytics server during that session.
ExitRates	For a specific page, the percentage of pageviews for which that page was the final pageview in the session.
PageValues	The average value of the pages visited by a user before completing an e-commerce transaction.

2.3.3 Timing and Seasonal Characteristics

The following variables describe when the shopping session occurred and whether it took place near a special shopping period.

Table 4. Timing and Seasonal Variables

Variable	Description
Month	Month in which the shopping session occurred.
SpecialDay	A measure of how close the visit occurred to a special shopping day, such as Valentine's Day or Mother's Day, when sessions may be more likely to result in a transaction. Ranges from 0 to 1, with higher values indicating that the visit occurred closer to the relevant special day.
Weekend	Boolean indicator of whether the shopping session occurred on a weekend.

2.3.4 Visitor and Technical Characteristics

These variables describe characteristics of the visitor, their browsing environment, and how they arrived at the website. The variables OperatingSystems, Browser, Region, and TrafficType are represented as numerical codes in the original dataset but represent categorical characteristics rather than continuous numerical measurements. These variables were therefore treated as categorical variables during preprocessing.

Table 5. Visitor and Technical Variables

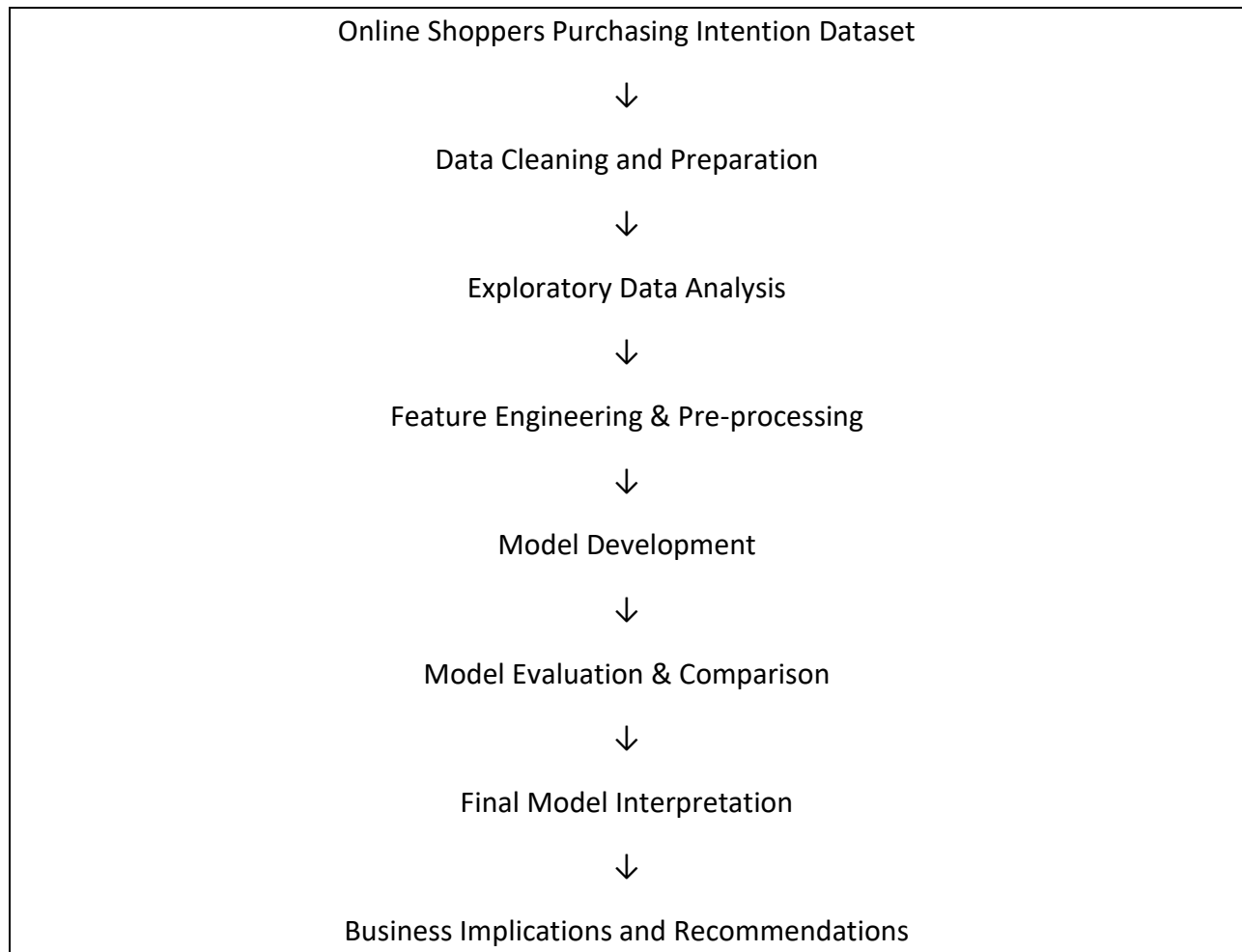
Variable	Description
VisitorType	Visitor classification indicating whether the visitor was a new visitor, returning visitor, or other.
OperatingSystems	Operating system used by the visitor.
Browser	Browser used by the visitor.
Region	Geographic region associated with the visitor.
TrafficType	Traffic source or type through which the visitor arrived at the website.

3.0 Methodology

This project followed a structured machine learning workflow, beginning with the raw online shopping session data and progressing through data cleaning, exploratory data analysis, preprocessing, model development, evaluation, and interpretation. The objective was to develop predictive models that could identify online shopping sessions likely to result in a purchase while comparing the performance of multiple machine learning algorithms.

The analysis focused on understanding patterns in visitor browsing behavior and session characteristics, preparing the data for predictive modeling, and evaluating how well different models could distinguish between purchasing and non-purchasing sessions. Figure 1 summarizes the overall analytical workflow used throughout the project.

Figure 1. Analytical Workflow



4.0 Data Cleaning and Preparation

The data cleaning process began with an initial inspection of the dataset to confirm its dimensions, variables, data types, and overall quality. The dataset contained 12,330 observations and 18 variables, including 17 predictor variables and the binary outcome variable, Revenue.

The inspection included reviewing the dataset structure, variable names, data types, and summary statistics. The dataset contained numerical, categorical, and Boolean variables, which informed the preprocessing approach used during model development.

4.1 Duplicate Observations

The dataset contained 125 completely duplicated rows, representing approximately 1% of all observations. Because the data represents individual online shopping sessions, identical combinations of the available variables may reflect legitimate separate sessions rather than data-entry or processing errors. These observations were therefore retained and included in subsequent analyses and modeling.

4.2 Missing Values

Missing values were examined across all variables before further analysis. The dataset did not contain any missing values, so no observations were removed and no imputation was required. This allowed the full dataset to be retained for exploratory analysis and predictive modeling.

4.3 Outlier Assessment

Potential outliers in the numerical variables were examined during exploratory data analysis using the $1.5 \times$ Inter-Quartile-Range (IQR) method. Several variables contained a substantial number of observations outside the calculated IQR thresholds, particularly PageValues, Informational, and Informational_Duration.

However, many of these variables were highly right-skewed or zero-inflated. For some variables, both the first and third quartiles were equal to zero, causing the IQR method to classify many nonzero observations as statistical outliers. Other variables contained observations in the upper tails of their distributions that appeared consistent with legitimate differences in visitor browsing and engagement behavior.

Because there was no evidence that these observations represented data-entry errors or invalid sessions, no observations were removed solely based on the IQR analysis. The full dataset was therefore retained for subsequent preprocessing and modeling.

Overall, the dataset required limited cleaning. No observations were removed or imputed, and the full dataset was retained for exploratory analysis and modeling. Additional preprocessing required for machine learning is described in Section 6.0.

5.0 Exploratory Data Analysis

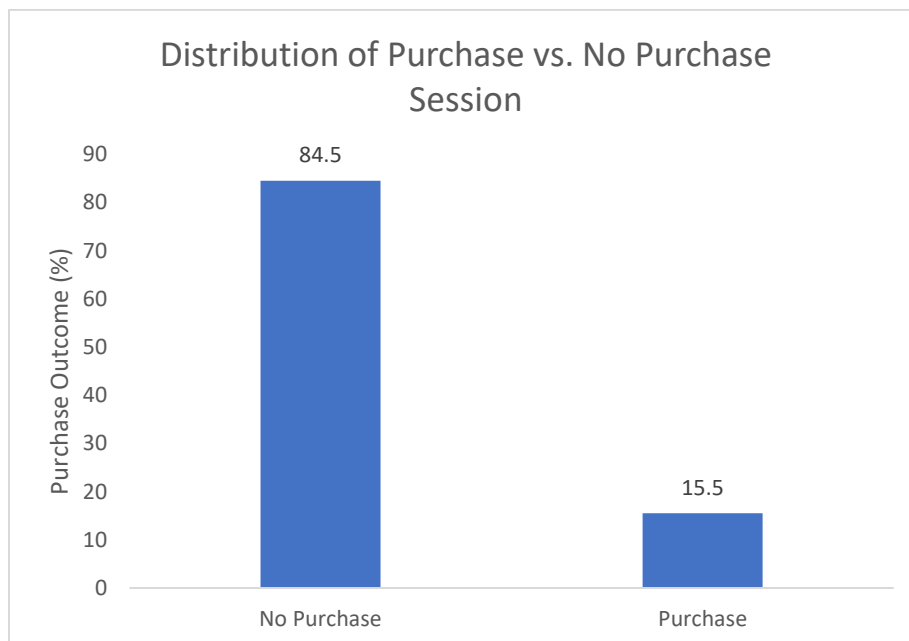
Exploratory Data Analysis (EDA) was conducted to examine the characteristics of online shopping sessions and identify patterns associated with purchase outcomes. The analysis included descriptive statistics and comparisons of browsing behavior, engagement, and session characteristics between purchasing and non-purchasing sessions.

Several numerical variables were highly right-skewed and contained a large proportion of zero values. These distributional characteristics were considered when interpreting the results and during subsequent preprocessing.

5.1 Purchase Distribution

The distribution of the outcome variable shows that most online shopping sessions did not result in a purchase. Approximately 15.5% of the 12,330 sessions resulted in a purchase, while approximately 84.5% did not.

Figure 2. Distribution of Purchase vs. No Purchase Sessions



This indicates a moderate class imbalance in the dataset. Although the majority of observations represent non-purchasing sessions, the dataset contains a sufficient number of purchasing sessions to support predictive modeling. However, the imbalance also means that accuracy alone is not sufficient for evaluating model performance. A model could achieve relatively high accuracy by correctly classifying the majority of non-purchasing sessions while failing to identify a meaningful proportion of actual purchasers.

For this reason, precision, recall, F1-score, and ROC-AUC were also considered during model evaluation.

5.2 Summary Statistics

The numerical variables provide an overview of how visitors interacted with the website during an online shopping session. Several variables, particularly the page visit and duration measures,

were highly right-skewed. Many sessions involved little or no activity within certain categories of pages, while a smaller number of sessions involved substantially greater browsing activity.

Variables such as Informational, Informational_Duration, and PageValues contained a large number of zero values, while measures of product-related activity showed greater variation across sessions. The distributions of BounceRates and ExitRates also varied substantially, reflecting differences in how visitors moved through and exited the website.

Table 6. Summary Statistics of Numerical Variables

Variable	Mean	25th Percentile	Median	75th Percentile	Maximum
Administrative	2.32	0.00	1.00	4.00	27.00
Administrative_Duration	80.82	0.00	7.50	93.26	3398.75
Informational	0.50	0.00	0.00	0.00	24.00
Informational_Duration	34.47	0.00	0.00	0.00	2549.38
ProductRelated	31.73	7.00	18.00	38.00	705.00
ProductRelated_Duration	1194.75	184.14	598.94	1464.16	63973.52
BounceRates	0.02	0.00	0.00	0.02	0.20
ExitRates	0.04	0.01	0.03	0.05	0.20
PageValues	5.89	0.00	0.00	0.00	361.76
SpecialDay	0.06	0.00	0.00	0.00	1.00

The categorical variables also showed uneven distributions across the dataset. May was the most frequently represented month, accounting for 3,364 sessions. Returning visitors represented the majority of observations, with 10,551 of the 12,330 sessions, while approximately 76.7% of sessions occurred on weekdays rather than weekends.

5.3 Key Relationships with Purchase Outcome

Several patterns emerged when comparing browsing behavior and session characteristics between purchasing and non-purchasing sessions. The clearest differences were observed in measures related to visitor engagement, particularly PageValues, product-related activity, and exit behavior.

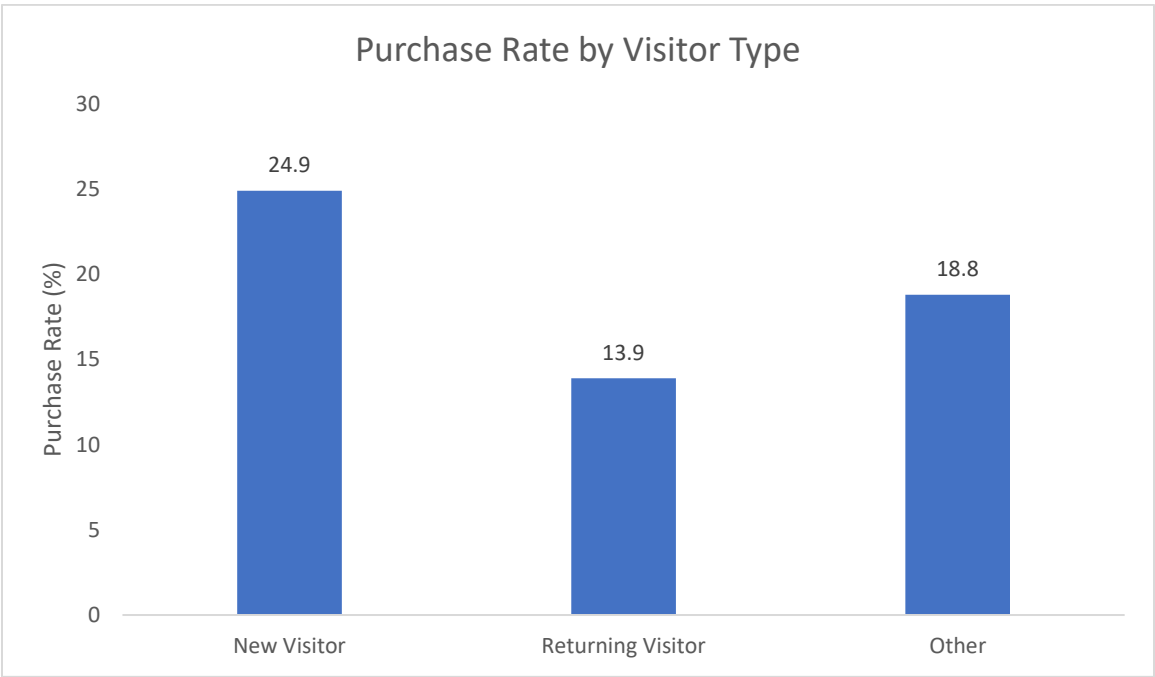
5.3.1 Purchase Outcome and Visitor Type

Visitor type provided useful information about differences in purchasing behavior across online shopping sessions. Figure 3 compares purchase rates across visitor types to examine whether new, returning, and other visitors demonstrated different patterns of purchase activity.

New visitors had the highest observed purchase rate in the dataset at 24.9%, compared with 18.8% among visitors classified as Other and 13.9% among returning visitors. Although returning visitors represented the majority of sessions in the dataset, they had the lowest observed purchase rate across the three visitor categories.

This pattern suggests that visitor type may provide useful information for predicting purchase intent, particularly in distinguishing between new and returning visitors.

Figure 3. Purchase Rate by Visitor Type



5.3.2 Purchase Outcome by Month

Purchase rates varied across the months represented in the dataset, suggesting that seasonal patterns may be associated with the likelihood that an online shopping session results in a transaction. Figure 4 shows the observed purchase rate for each month included in the dataset.

November had the highest observed purchase rate at 25.4%, followed by October (20.9%) and September (19.2%). In contrast, February had the lowest purchase rate at 1.6%. Purchase rates generally increased during the later months of the year before declining again in December (12.5%).

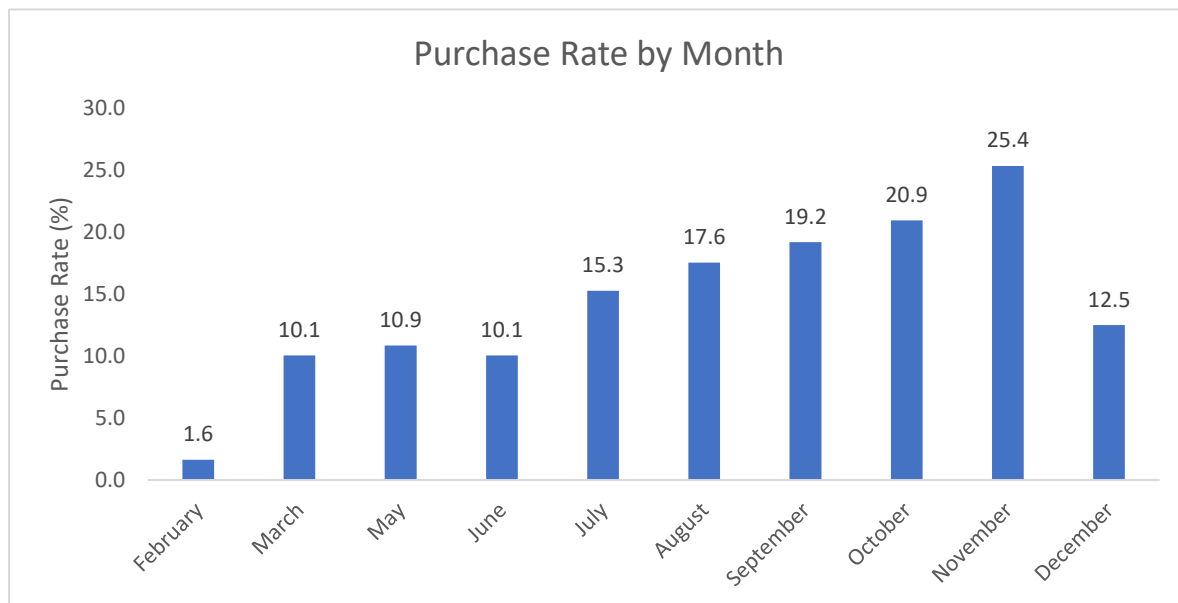
The higher purchase rates observed during the fall and early holiday season may reflect broader seasonal patterns in consumer spending in the United States. November includes major shopping periods such as Thanksgiving and Black Friday, while December coincides with the

Christmas and year-end holiday season. These periods are often associated with increased consumer spending and promotional activity, which may contribute to higher levels of online purchasing.

The decline in December compared with November also suggests that seasonal patterns are not uniform across the entire holiday period. However, because the dataset does not contain observations for January or April, it is not possible to examine whether purchase activity changed further following the holiday season.

The variation in purchase rates across months supports the inclusion of Month as a predictor in the machine learning models. Seasonal timing may provide useful information about purchase intent, although it represents only one aspect of an online shopping session alongside more direct measures of visitor behavior and website engagement.

Figure 4. Purchase Rate by Month



5.3.3 Purchase Outcome by Day of the Week

Purchase rates were slightly higher for sessions that occurred during the weekend compared with those that occurred on weekdays. As shown in Table 7, 17.4% of weekend sessions resulted in a purchase, compared with 14.9% of weekday sessions. While the difference was relatively modest, it suggests that the timing of a shopping session may provide some additional information about purchase intent. This supports the inclusion of the Weekend variable in the predictive models, alongside other measures of visitor behavior and engagement.

Table 7. Purchase Rate by Weekend/Weekday

Day of the Week	Purchase Rate (%)
Weekend	17.4
Weekday	14.9

5.3.4 Purchase Outcome and Browsing/Engagement Behaviors

Table 8 compares the median values of selected browsing and engagement variables for purchasing and non-purchasing sessions. Several clear differences were observed between the two groups.

Table 8. Purchase Outcome and Browsing/Engagement Behaviors

Variable	Non-Purchasing Session	Purchasing Session
Administrative	0.00	2.00
Administrative_Duration	0.00	52.37
Informational	0.00	0.00
Informational_Duration	0.00	0.00
ProductRelated	16.00	29.00
ProductRelated_Duration	510.19	1109.91
BounceRates	0.00	0.00
ExitRates	0.03	0.02
PageValues	0.00	16.76
SpecialDay	0.00	0.00

Purchasing sessions showed substantially greater engagement with product-related content. The median number of product-related pages visited was 29 among purchasing sessions, compared with 16 among non-purchasing sessions. Purchasing sessions also spent more than twice as much time on product-related pages, with a median duration of approximately 1,109.9 seconds, compared with 510.2 seconds for non-purchasing sessions.

Purchasing sessions also showed greater activity on administrative pages. The median number of administrative pages visited was 2, compared with 0 for non-purchasing sessions, while the median time spent on administrative pages was approximately 52.4 seconds for purchasing sessions, compared with 0 seconds for non-purchasing sessions.

Small differences were also observed in website engagement metrics. Purchasing sessions had lower median bounce rates (0.0000 vs. 0.0043) and lower median exit rates (0.0160 vs. 0.0286)

than non-purchasing sessions. This suggests that sessions resulting in a purchase were generally associated with lower bounce and exit behavior.

The largest difference was observed for PageValues. The median PageValues was 0 for non-purchasing sessions and approximately 16.8 for purchasing sessions. This suggests a particularly strong relationship between PageValues and the purchase outcome, although this variable should be interpreted carefully because of how it is defined in relation to pages visited before an e-commerce transaction.

Overall, the exploratory analysis suggests that purchasing sessions were generally characterized by deeper engagement with the website, including more product-related browsing, more time spent on product pages, greater administrative activity, and lower bounce and exit behavior. These patterns provided an initial indication that measures of browsing behavior and website engagement could be useful predictors in the machine learning models.

5.4 Key Insights from EDA

Overall, purchasing sessions were associated with deeper website engagement, including greater product-related and administrative activity, lower exit behavior, and substantially higher PageValues. Visitor type and timing also showed differences in purchase rates. These patterns supported the inclusion of behavioral, engagement, temporal, and visitor characteristics in the predictive models.

6.0 Feature Engineering and Pre-processing

Before model development, the predictor variables (X) were separated from the outcome variable (y). The dataset was then divided into 80% training data (9,864 observations) and 20% test data (2,466 observations) using stratified sampling to preserve the proportion of purchasing and non-purchasing sessions in both subsets.

The 17 original predictor variables included numerical, categorical, and Boolean features. A preprocessing pipeline using a ColumnTransformer was created to apply the appropriate transformations. The 10 numerical variables were standardized, the six categorical variables were one-hot encoded, and the Boolean Weekend variable was retained as a binary feature.

The preprocessing transformer was fit only on the training data and then applied to the test data. This helped prevent data leakage by ensuring that information from the held-out test dataset was not used when calculating scaling parameters or defining the encoded feature structure.

Following preprocessing, the 17 original variables were expanded into 74 model-ready features, primarily because of one-hot encoding. A separate preprocessing approach was used for the deep neural network models and is described in the model development section.

6.1 Consideration of PageValues

During EDA, PageValues was identified as a potentially important predictor of purchase outcome. The dataset documentation defines it as the average value of pages visited by a user before completing an e-commerce transaction, raising the possibility of temporal leakage.

PageValues was retained because it is included as a predictor in the original dataset and has been used in previous analyses of purchase behavior. However, the timing of when this information becomes available during a shopping session may limit its usefulness for early or real-time intervention. This limitation is considered when interpreting the final model and its potential business applications.

7.0 Model Development

Three machine learning approaches were developed to predict whether an online shopping session would result in a purchase: Logistic Regression, Random Forest, and Deep Neural Networks. Logistic Regression served as the baseline model, while Random Forest and Deep Neural Networks were used to evaluate whether more flexible and non-linear approaches could improve predictive performance.

Initial and tuned versions of the Random Forest and Deep Neural Network models were evaluated. All models were assessed on the test dataset using accuracy, precision, recall, F1-score, and ROC-AUC. Because purchasing sessions represented a minority of observations, model selection considered the balance across these metrics rather than accuracy alone.

7.1 Logistic Regression

Logistic Regression was selected as the baseline model because it is a widely used and interpretable approach for binary classification problems. It provides a useful benchmark against which the performance of the more flexible Random Forest and Deep Neural Network models can be compared.

The model achieved an accuracy of 88.1% and a ROC-AUC of 0.883, indicating a relatively strong ability to distinguish between purchasing and non-purchasing sessions. The model also achieved a precision of 74.1%, meaning that when it predicted a session would result in a purchase, the prediction was often correct.

However, recall for purchasing sessions was substantially lower at 35.9%. This indicates that while the model was relatively precise when identifying likely purchasers, it captured only about one-third of the sessions that actually resulted in a purchase. The classification results also showed a clear difference in performance between the two outcome classes: recall was approximately 98% for non-purchasing sessions, compared with 36% for purchasing sessions. This difference reflects, in part, the imbalance in the outcome variable, with non-purchasing sessions representing the majority of the dataset.

The combination of a relatively strong ROC-AUC and lower recall at the default classification threshold suggests that the model contained useful predictive information but was relatively conservative when assigning sessions to the purchasing class. Overall, Logistic Regression provided a useful and interpretable baseline, while also highlighting the potential value of more flexible models that may be better able to identify purchasing sessions.

Table 9. Logistic Regression Performance

Metric	Value
Accuracy	0.881
Precision	0.741
Recall	0.359
F1 Score	0.483
ROC-AUC	0.883

7.2 Random Forest (Baseline and Tuned)

Random Forest was included as a nonlinear machine learning model to compare with the Logistic Regression baseline and Deep Neural Network models. Unlike Logistic Regression, Random Forest can capture more complex relationships and interactions among features without requiring them to be specified in advance. A baseline model was first evaluated using the default settings, followed by hyperparameter tuning using RandomizedSearchCV with three-fold cross-validation on the training data. The search evaluated 10 parameter combinations and used F1-score as the optimization metric because of the imbalanced outcome variable. The held-out test dataset was not used during tuning and was reserved for final model evaluation.

7.2.1 Baseline Random Forest

The Baseline Random Forest outperformed Logistic Regression across all five evaluation metrics. It achieved 90.0% accuracy, 76.6% precision, 51.3% recall, an F1-score of 0.614, and a ROC-AUC of 0.918. The largest improvement was in recall for purchasing sessions, which

increased from 35.9% with Logistic Regression to 51.3%, while precision also improved. These results suggest that the Random Forest was better able to capture relationships in the data that helped distinguish purchasing from non-purchasing sessions.

7.2.2 Tuned Random Forest

Hyperparameter tuning further improved the model's ability to identify purchasing sessions. Recall increased from 51.3% to 68.3%, while the F1-score increased from 0.614 to 0.663 and ROC-AUC increased slightly to 0.923. This improvement came with a trade-off: precision decreased from 76.6% to 64.4%, and accuracy decreased slightly from 90.0% to 89.3%. The best-performing configuration included `class_weight="balanced"`, suggesting that accounting for the class imbalance helped the model identify a greater proportion of actual purchasers.

Overall, the Baseline Random Forest achieved higher accuracy and precision, while the Tuned Random Forest achieved substantially higher recall, F1-score, and ROC-AUC. Given the project's objective of identifying sessions likely to result in a purchase, the Tuned Random Forest provided the stronger balance between identifying actual purchasers and maintaining overall predictive performance.

Table 10. Random Forest Model Performance

Metric	Baseline Random Forest	Tuned Random Forest
Accuracy	0.900	0.893
Precision	0.766	0.644
Recall	0.513	0.683
F1 Score	0.614	0.663
ROC-AUC	0.918	0.923

7.3 Deep Neural Networks (Initial and Tuned)

Deep neural networks were included to evaluate whether a more flexible model could capture complex relationships between browsing behavior, website engagement, visitor characteristics, and purchase outcomes. The training data was further divided into training and validation subsets using stratified sampling. Preprocessing was fit only on the neural network training data and then applied to the validation and held-out test datasets. Early stopping was used to monitor validation performance and reduce overfitting.

7.3.1 Initial Deep Neural Network

The initial deep neural network used two fully connected hidden layers containing 64 and 32 neurons, respectively, with ReLU activation functions. Dropout layers were included to reduce the risk of overfitting, and the output layer used a sigmoid activation function to estimate the probability that a session would result in a purchase.

The model completed 15 of the maximum 50 training epochs before early stopping was triggered. Validation performance improved substantially during the early stages of training, with the lowest validation loss occurring around Epoch 10. After this point, training loss continued to decrease while validation loss fluctuated, suggesting the beginning of mild overfitting. The best model weights were restored based on validation performance.

On the held-out test dataset, the initial DNN achieved an accuracy of 89.3%, precision of 69.6%, recall of 55.2%, an F1-score of 0.616, and a ROC-AUC of 0.911. The model substantially improved recall compared with Logistic Regression, although it did not outperform the Tuned Random Forest in recall, F1-score, or ROC-AUC.

7.3.2 Tuned Deep Neural Network

To evaluate whether additional model capacity could improve performance, the neural network architecture was modified by increasing the number of neurons in the two hidden layers from $64 \rightarrow 32 \rightarrow 1$ to $128 \rightarrow 64 \rightarrow 1$. The tuned model retained the same two-layer structure, activation functions, dropout rate, optimizer, loss function, batch size, validation strategy, and early stopping procedure as the initial model. This allowed the effect of increasing the number of neurons to be evaluated without introducing additional changes to the model.

The tuned network used 74 processed input features, followed by a hidden layer with 128 neurons, a dropout layer with a rate of 0.20, a second hidden layer with 64 neurons, and another dropout layer with a rate of 0.20. The final output layer contained one neuron with a sigmoid activation function.

The tuned model completed 11 of the maximum 50 training epochs before early stopping was triggered. Validation loss reached its lowest value at Epoch 6 and then fluctuated or worsened, while training loss continued to decrease. This pattern again suggested the beginning of mild overfitting. Validation accuracy and recall improved during the early stages of training but plateaued relatively quickly, suggesting that the additional model complexity did not result in a clear improvement in validation performance.

On the held-out test dataset, the tuned DNN achieved an accuracy of 89.5%, precision of 70.9%, recall of 54.2%, an F1-score of 0.614, and a ROC-AUC of 0.912. Compared with the initial model,

the tuned network achieved slightly higher accuracy, precision, and ROC-AUC, while the initial model achieved slightly higher recall and F1-score.

Table 11. Deep Neural Network Performance Comparison

Metric	Initial Deep Neural Network	Tuned Deep Neural Network
Accuracy	0.893	0.895
Precision	0.696	0.709
Recall	0.552	0.542
F1 Score	0.616	0.614
ROC-AUC	0.911	0.912

Overall, increasing the size of the hidden layers produced only marginal differences in performance. Both neural network models demonstrated similar predictive ability, but neither outperformed the Tuned Random Forest, which achieved higher recall, F1-score, and ROC-AUC. The results suggest that increasing model complexity alone did not provide a substantial performance benefit for this dataset.

7.4 Model Comparison

Table 12 compares the performance of all five models on the held-out test dataset. The Tuned Random Forest achieved the highest F1-score (0.663) and ROC-AUC (0.923) of all models, as well as the highest recall for purchasing sessions (68.3%). Although the Baseline Random Forest achieved the highest accuracy (90.0%) and precision (76.6%), it identified substantially fewer purchasing sessions than the tuned model. The Logistic Regression model had the lowest recall and F1-score, while the two deep neural network models produced similar results but did not outperform the Tuned Random Forest on the key evaluation metrics. Based on the project objective of identifying sessions likely to result in a purchase, while maintaining a reasonable balance between precision and recall, the Tuned Random Forest was selected as the final model.

Table 12. Model Comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic regression	0.881	0.741	0.359	0.483	0.883
Baseline Random Forest	0.900	0.766	0.513	0.614	0.918
Tuned Random Forest	0.893	0.644	0.683	0.663	0.923
Initial Deep Neural Network	0.893	0.696	0.552	0.616	0.911

Tuned Deep Neural Network	0.895	0.709	0.542	0.614	0.912
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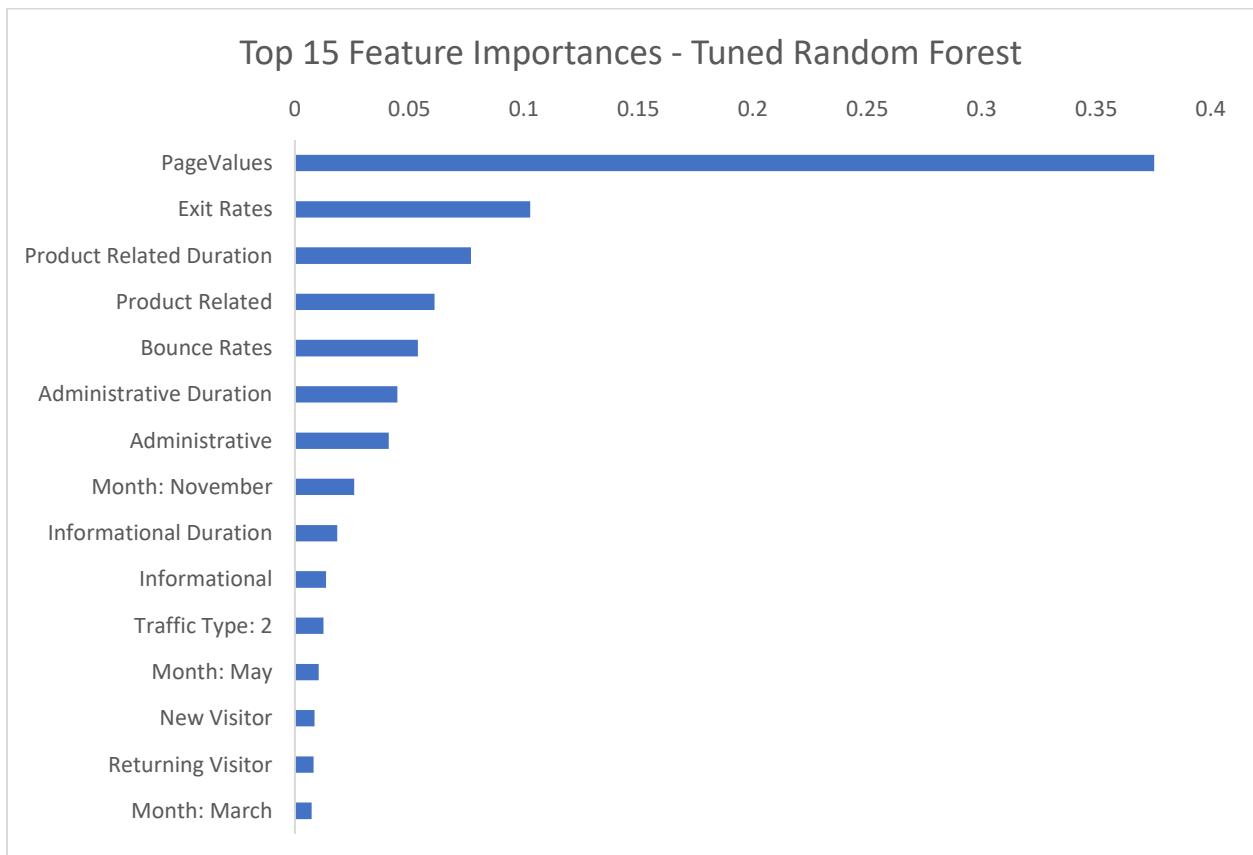
8.0 Final Model and Interpretation

The Tuned Random Forest was selected as the final model based on its ability to identify purchasing sessions while maintaining strong overall predictive performance. Feature importance was examined to better understand which characteristics of an online shopping session contributed most to the model's predictions.

8.1 Feature Importance

Feature importance was examined to better understand which characteristics of an online shopping session contributed most to the Tuned Random Forest's predictions. The results were consistent with several of the patterns identified during exploratory data analysis: measures of browsing behavior and website engagement were substantially more important to the model than temporal or visitor classification characteristics.

Figure 5. Feature Importance - Tuned Random Forest



PageValues was by far the most important feature, followed by ExitRates, product-related page activity, BounceRates, and administrative activity. These results are consistent with the EDA findings and suggest that how visitors engage with the website provides substantially more predictive information than broader temporal or visitor characteristics.

Month, traffic source, and visitor type appeared among the top features but had considerably lower importance. These results should be interpreted as indicators of predictive usefulness rather than causality and do not indicate whether a feature increases or decreases the likelihood of purchase. The importance of PageValues should also be interpreted carefully because of its relationship to transaction activity and potential limitations for real-time prediction.

9.0 Business Implications

The results suggest that information about visitor behavior during an online shopping session could potentially be used to identify sessions with a higher likelihood of resulting in a purchase. Engagement with product-related pages, exit behavior, and other measures of website activity provided particularly useful predictive information.

In practice, predictions could potentially support targeted promotions, product recommendations, marketing prioritization, or improvements to website design and user experience. However, practical implementation would require additional validation and careful consideration of when key features, particularly PageValues, become available during a session.

10.0 Study Limitations

This study has several limitations that should be considered when interpreting the results:

- **Single dataset:** The analysis was based on online shopping sessions from a single dataset, which may limit how well the findings generalize to other websites, industries, or customer populations.
- **Historical data:** The dataset reflects online shopping behavior during a specific period and may not fully represent current consumer behavior or website technologies.
- **Predictive, not causal:** The analysis identifies characteristics associated with purchase outcomes but does not establish that these factors cause a visitor to make a purchase.
- **Feature availability:** PageValues was the most important predictor in the final model, but its usefulness in a real-time setting may depend on when the metric becomes available during a shopping session.

- **Limited information:** The dataset does not include other potentially useful factors, such as customer purchase history, product characteristics, pricing, or individual-level demographic information.

11.0 Recommendations

Based on the results of this analysis, the following recommendations are suggested:

- **Prioritize behavioral engagement metrics** when predicting purchase intent, as features related to page engagement, exit behavior, and product-related activity were among the most important predictors.
- **Consider the Tuned Random Forest model** when the goal is to identify a larger proportion of potential purchasers while maintaining a reasonable balance between precision and recall.
- **Evaluate the timing and availability of key features**, particularly PageValues, before applying the model in a real-time setting.
- **Validate the model using website-specific and more recent data** before considering practical implementation.

12.0 Future Work

Several opportunities for future work could build on the findings of this project:

- **Evaluate the model without PageValues** to assess how strongly overall performance depends on this feature and whether the model could be more suitable for real-time prediction.
- **Explore additional machine learning models and tuning strategies**, including gradient boosting methods and more extensive hyperparameter optimization.
- **Evaluate different classification thresholds** based on the relative costs of missing potential purchasers versus incorrectly identifying non-purchasing sessions as likely purchasers.
- **Validate the model using more recent or website-specific data** to assess how well the findings generalize to different online shopping environments.
- **Incorporate additional information**, such as customer purchase history, product characteristics, pricing, or other behavioral data, where available.